

POWER BI – DATA IMPORT & TRANSFORMATION PROJECT REPORT

Project Overview: This project focuses on transforming raw business data into meaningful insights using Power **Query** and **Power BI**. Three datasets—List of Orders, Order Details, and Sales Target—were cleaned, formatted, merged, and modelled to create a structured analytical dataset. The project demonstrates essential data preparation techniques such as data type correction, text formatting, column creation, conditional logic, grouping, aggregation, and relationship modelling. The final output enables accurate reporting, state-level filtering, category-wise profit analysis, and monthly target comparison.

Tools & Technologies Used:

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Basic Calculations)
- CSV Data Files Business

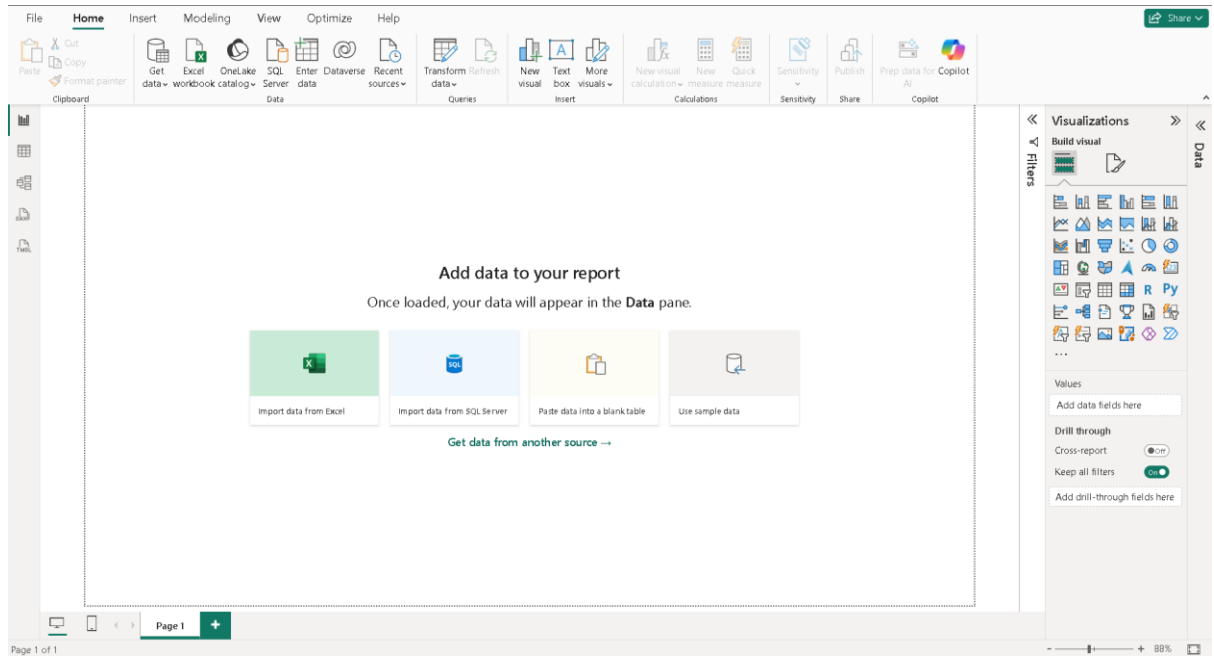
Problem Statement:

Businesses often struggle to analyze sales performance due to inconsistent data formats, missing values, and unstructured datasets. This project aims to:

- Clean and standardize order data
- Combine multiple datasets into a single analytical model
- Identify profit trends across categories
- Compare actual performance with monthly targets
- Enable state-level and category-level insights The goal is to create a reliable dataset for dashboards and decision-making.

1. Data Import:

- Imported List of Orders.csv, Order Details.csv, and Sales Target.csv into Power BI.



Opened all datasets in Power Query Editor using Transform Data for cleaning and shaping.

Order ID	Order Date	Customer Name	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Ilitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavina	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	23-04-2018	Vini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bilalim	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna
29	24-04-2018	Hiemant	Kerala	Thiruvananthapuram
30	24-04-2018	Sahil	Punjab	Chandigarh

2. Row Limiting & Data Types:

- Limited List of Orders to the first 500 rows using Keep Top Rows

×

Keep Top Rows

Specify how many rows to keep.

Number of rows

500

OK

Cancel

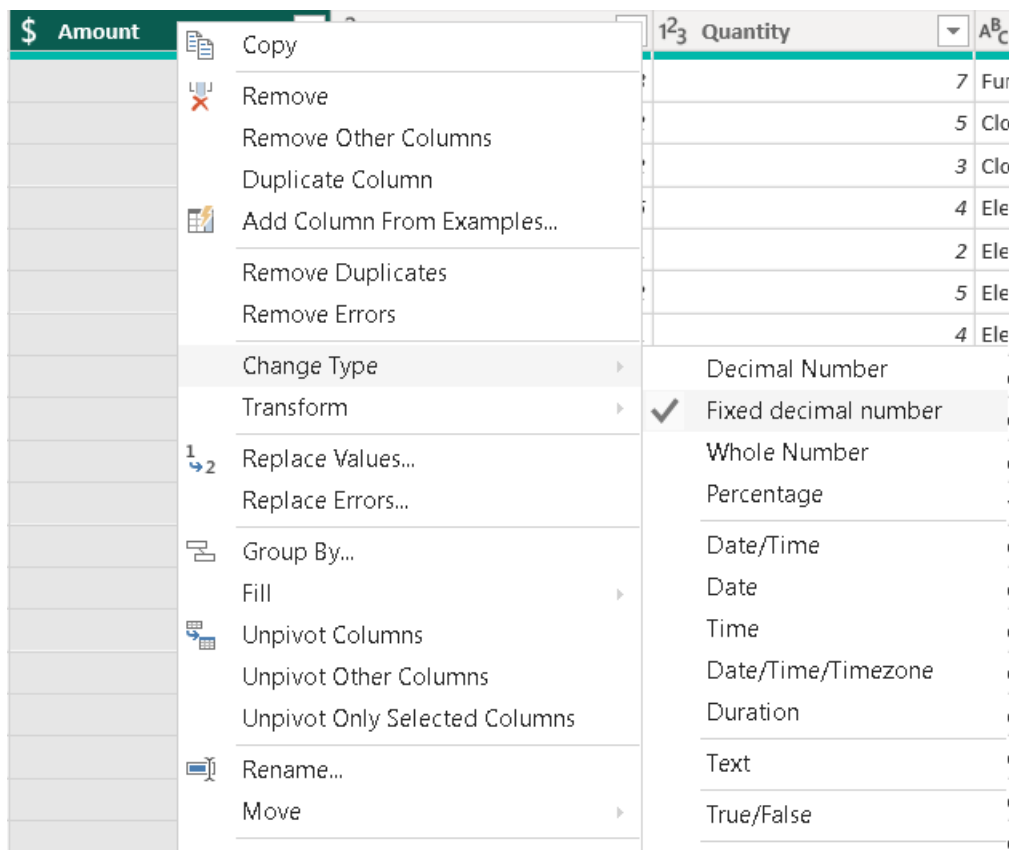
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Kept First Rows**

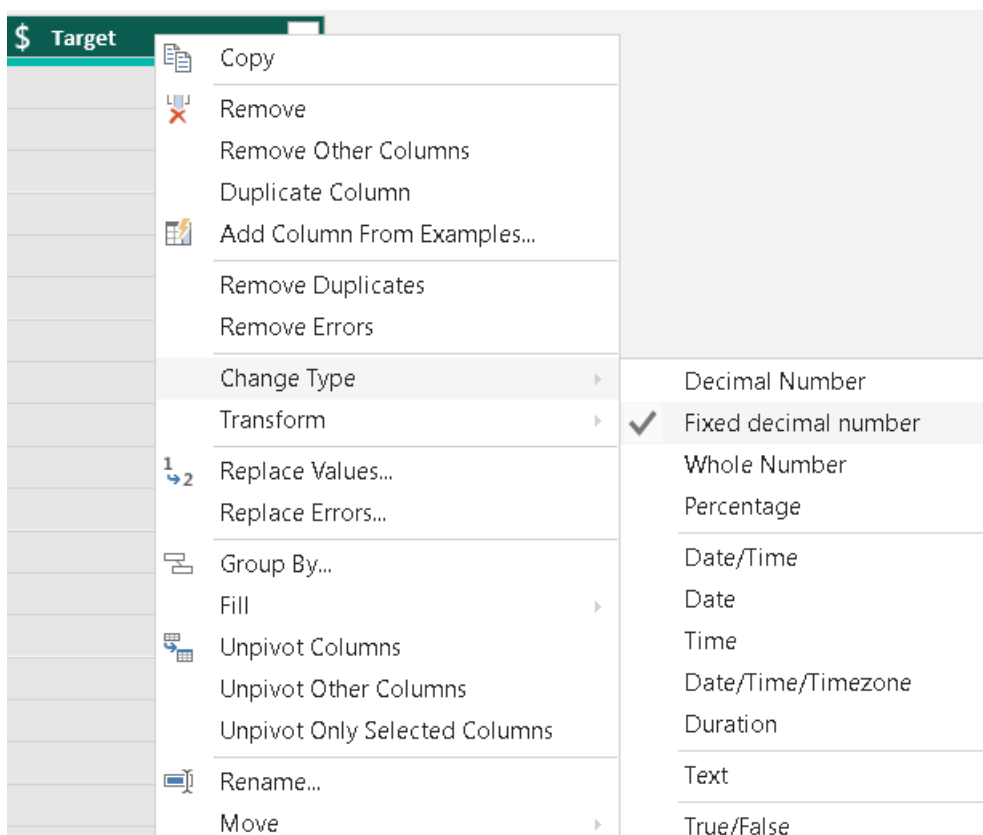
Converted: Order Date → Date type

Order Date	Copy	AB State	AB C
	Remove	Gujarat	Ahme
	Remove Other Columns	Maharashtra	Pune
	Duplicate Column	Madhya Pradesh	Bhopa
	Add Column From Examples...	Rajasthan	Jaipur
	Remove Duplicates	West Bengal	Kolkat
	Remove Errors	Karnataka	Banga
	Change Type	Jammu and Kashmir	Kashm
	Transform		
1 2	Replace Values...		
	Replace Errors...		
	Group By...		
	Fill		
	Unpivot Columns		
	Unpivot Other Columns		
	Unpivot Only Selected Columns		
	Rename...		
	Move		
	Drill Down		
	Add as New Query		

- Amount → Fixed Decimal Number



- Target → Fixed Decimal Number



3. Text Formatting:

- Standardized customer names by applying Proper Case to the Customer Name column for consistent formatting

The screenshot shows the Power Query Editor interface. On the left, a table with columns 'CustomerName' and 'City' is visible. The 'CustomerName' column contains names like Bharat, Pearl, Jahan, Divsha, Kasheen, Hazel, Sonakshi, Aarushi, Jitesh, Yogesh, Anita, Shrichand, Mukesh, Vandana, and Bhavna. The 'City' column contains city names like Ahmedabad, Pune, Bhopal, Jaipur, Kolkata, Bangalore, Kashmir, and Chennai. A context menu is open over the 'CustomerName' column, with the 'Transform' option selected. The 'Transform' submenu is also open, showing options like lowercase, UPPERCASE, Capitalize Each Word (which is highlighted), Trim, Clean, Length, JSON, and XML.

CustomerName	City
Bharat	Ahmedabad
Pearl	Pune
Jahan	Bhopal
Divsha	Jaipur
Kasheen	Kolkata
Hazel	Bangalore
Sonakshi	Kashmir
Aarushi	Chennai
Jitesh	
Yogesh	
Anita	
Shrichand	
Mukesh	
Vandana	
Bhavna	

The screenshot shows the 'APPLIED STEPS' pane in the Power Query Editor. It lists the steps applied to the data: Source, Promoted Headers, Changed Type, Kept First Rows, Changed Type1, and Capitalized Each Word. The 'Capitalized Each Word' step is currently selected and highlighted.

APPLIED STEPS	
Source	⚙️
Promoted Headers	⚙️
Changed Type	
Kept First Rows	⚙️
Changed Type1	
Capitalized Each Word	

4. Column Creation:

○ Merge the City and State columns to create a new column named Location in the format: City, State

CustomerName	State	City
arat	Gujarat	Ahmedabad
arl	Maharashtra	Pune
an	Madhya Pradesh	Bhopal

Merge Columns

Choose how to merge the selected columns.

Separator
Comma

New column name (optional)
Location

OK Cancel

Location
Ahmedabad,Gujarat
Pune,Maharashtra
Bhopal,Madhya Pradesh
Jaipur,Rajasthan
Kolkata,West Bengal
Bangalore,Karnataka
Kashmir,Jammu and Kashmir
Chennai,Tamil Nadu
Lucknow,Uttar Pradesh
Patna,Bihar
Thiruvananthapuram,Kerala
Chandigarh,Punjab
Chandigarh,Haryana

APPLIED STEPS	
Source	
Promoted Headers	
Changed Type	
Kept First Rows	
Changed Type1	
Capitalized Each Word	
✕ Inserted Merged Column	

4.2 Profit Margin Column:

- Created a custom column: Profit Margin = Profit / Amount & Formatted as Percentage in Power BI.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ⓘ

= [Profit]/[Amount]

Available columns

Order ID

Amount

Profit

Quantity

Category

Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

% Profi Margin
-90.04%
-18.18%
-25.00%
-70.00%
-66.07%
-64.15%
43.98%
37.79%
-4.20%
-4.43%
-125.00%
-86.01%
2.78%
13.79%
33.64%
8.33%
47.37%
26.15%
3.18%
0.00%

APPLIED STEPS	
Source	⚙
Promoted Headers	⚙
Changed Type	
Added Custom	⚙
✕ Changed Type1	

ABC123

Profit Status

▼

LOSS

LOSS

LOSS

Profit

Profit

Profit

Profit

Profit

Profit

Profit

Profit

Break - Even

▲

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Custom

Changed Type1

✕ Added Conditional Column

×

Merge

Select tables and matching columns to create a merged table.

List of Orders

📄

Order ID	Order Date	CustomerName	State	City	Location
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal

Order Details

📄

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profi Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	LOSS
B-25601	66.00	-12	5	Clothing	Stole	-18.18%	LOSS
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00%	LOSS
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00%	LOSS
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	LOSS

Join Kind

Left Outer (all from first, matching from second)

▼

☒ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

- Named the merged output table Orders Data.
- Expanded required fields such as Amount, Profit, Category, etc.

The screenshot shows the Power BI Desktop interface. The main view displays a table with the following columns: Order Date, CustomerName, State, City, Location, and Order Details. The table contains 29 rows of data. A 'Search Columns to Expand' dialog is open on the right side of the screen. The dialog has a search bar and a list of columns to expand. The 'Expand' radio button is selected, and the 'Use original column name as prefix' checkbox is checked. The columns listed in the dialog are: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, and Profit Status. The 'OK' button is highlighted.

7. Handling Missing & Duplicate Data:

- Checked all tables for:

o Missing values

o Duplicate rows

The screenshot shows the Power BI Desktop interface. The main view displays a table with the following columns: Month of Order Date, Category, and Target. The table contains 26 rows of data. A 'Query Settings' panel is open on the right side of the screen. The 'APPLIED STEPS' section is expanded, and 'Removed Duplicates' is selected. The 'Properties' section shows the name 'Sales target' and 'All Properties'.

File	Home	Transform	Add Column	View	Tools	Help
Query Settings	Formula Bar	Monospaced	Column distribution	Always allow	Advanced Editor	Query Dependencies
Layout	Data Preview	Columns	Parameters	Advanced	Dependencies	

Queries [4]	Table.Distinct("Inserted Merged Column")
-------------	--

Order ID	Order Date	CustomerName	State	City	Location
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3 B-25603	03-04-2018	Iahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9 B-25609	09-04-2018	Ilitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10 B-25610	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15 B-25615	15-04-2018	Bhavina	Sikkim	Gangtok	Gangtok,Sikkim
16 B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17 B-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20 B-25620	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
21 B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
22 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23 B-25623	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24 B-25624	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25 B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26 B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27 B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 05:55

8. Sorting & Filtering:

- Sorted Orders Data by Order Date (Descending).
- Applied filters to analyze orders from Tamil Nadu for state-level insights.

File	Home	Transform	Add Column	View	Tools	Help
Query Settings	Formula Bar	Monospaced	Column distribution	Always allow	Advanced Editor	Query Dependencies
Layout	Data Preview	Columns	Parameters	Advanced	Dependencies	

Queries [4]	Table.SelectRows("Sorted Rows", each ([State] = "Tamil Nadu"))
-------------	--

Order ID	Order Date	CustomerName	State	City	Location	Amount
1 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
2 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
3 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
4 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
5 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
6 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
7 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
8 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
9 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
10 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
11 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
12 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
13 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
14 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
15 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
16 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
17 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
18 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
19 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
20 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
21 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
22 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
23 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
24 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
25 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
26 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
27 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	

13 COLUMNS, 271 ROWS Column profiling based on top 1000 rows

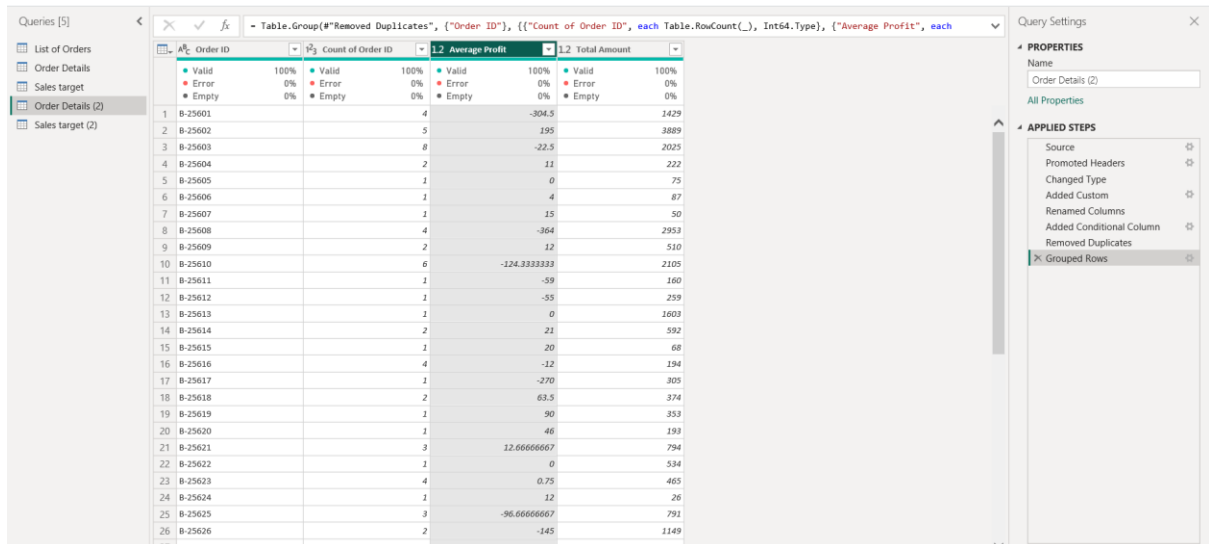
PREVIEW DOWNLOADED AT 05:58

9. Grouping & Aggregation:

9.1 Order Details Aggregations

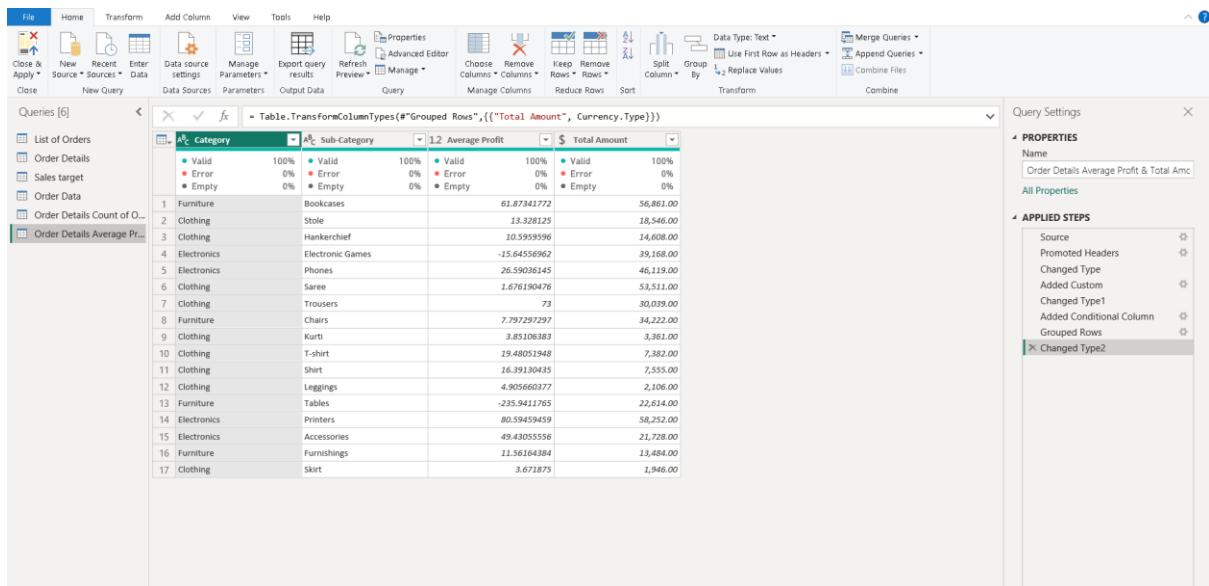
Duplicated Order Details and performed:

- Count of Order ID



Order ID	Count of Order ID	Average Profit	Total Amount
B-25601	4	-304.5	1429
B-25602	5	195	3889
B-25603	8	-22.5	2025
B-25604	2	11	222
B-25605	1	0	75
B-25606	1	4	87
B-25607	1	15	50
B-25608	4	-364	2953
B-25609	2	12	510
B-25610	6	-124.3333333	2105
B-25611	1	-59	160
B-25612	1	-55	259
B-25613	1	0	1603
B-25614	2	21	592
B-25615	1	20	68
B-25616	4	-12	194
B-25617	1	-270	305
B-25618	2	63.5	374
B-25619	1	90	353
B-25620	1	46	193
B-25621	3	12.66666667	794
B-25622	1	0	534
B-25623	4	0.75	465
B-25624	1	12	26
B-25625	3	-96.66666667	791
B-25626	2	-145	1149
B-25627	1	-38	55

- Average Profit & Total Amount



Category	Sub-Category	Average Profit	Total Amount
Furniture	Bookcases	61.87341772	56,861.00
Clothing	Stole	13.328125	18,546.00
Clothing	Hankkerchief	10.59595956	14,608.00
Electronics	Electronic Games	-15.64556962	39,168.00
Electronics	Phones	26.59036145	46,119.00
Clothing	Saree	1.676190476	53,511.00
Clothing	Trousers	73	30,039.00
Furniture	Chairs	7.797287297	34,222.00
Clothing	Kurti	3.85106383	3,361.00
Clothing	T-shirt	19.48051948	7,382.00
Clothing	Short	16.39130425	7,555.00
Clothing	Leggings	4.805680377	2,106.00
Furniture	Tables	-235.9411765	22,614.00
Electronics	Printers	80.59459459	58,252.00
Electronics	Accessories	49.43055556	21,728.00
Furniture	Furnishings	11.56164384	13,484.00
Clothing	Skirt	3.671875	1,946.00

9.2 Sales Target Aggregation:

- Duplicated Sales Target
- Added Month of Order Date column
- Grouped by Month to calculate Total Target per Month

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a table with two columns: 'Month of Order Date' and 'Total Target'. The table contains 12 rows of data, representing monthly sales targets from April 2018 to March 2019. The 'Total Target' values range from 31,400.00 to 43,800.00. The interface includes a ribbon at the top with various data transformation options, a left sidebar with a list of queries, and a right sidebar with query settings and applied steps.

	Month of Order Date	Total Target
1	01-04-2018	31,400.00
2	01-05-2018	31,500.00
3	01-06-2018	31,600.00
4	01-07-2018	33,800.00
5	01-08-2018	33,900.00
6	01-09-2018	34,000.00
7	01-10-2018	36,100.00
8	01-11-2018	36,300.00
9	01-12-2018	36,400.00
10	01-01-2019	43,500.00
11	01-02-2019	43,600.00
12	01-03-2019	43,800.00

10. Data Modelling:

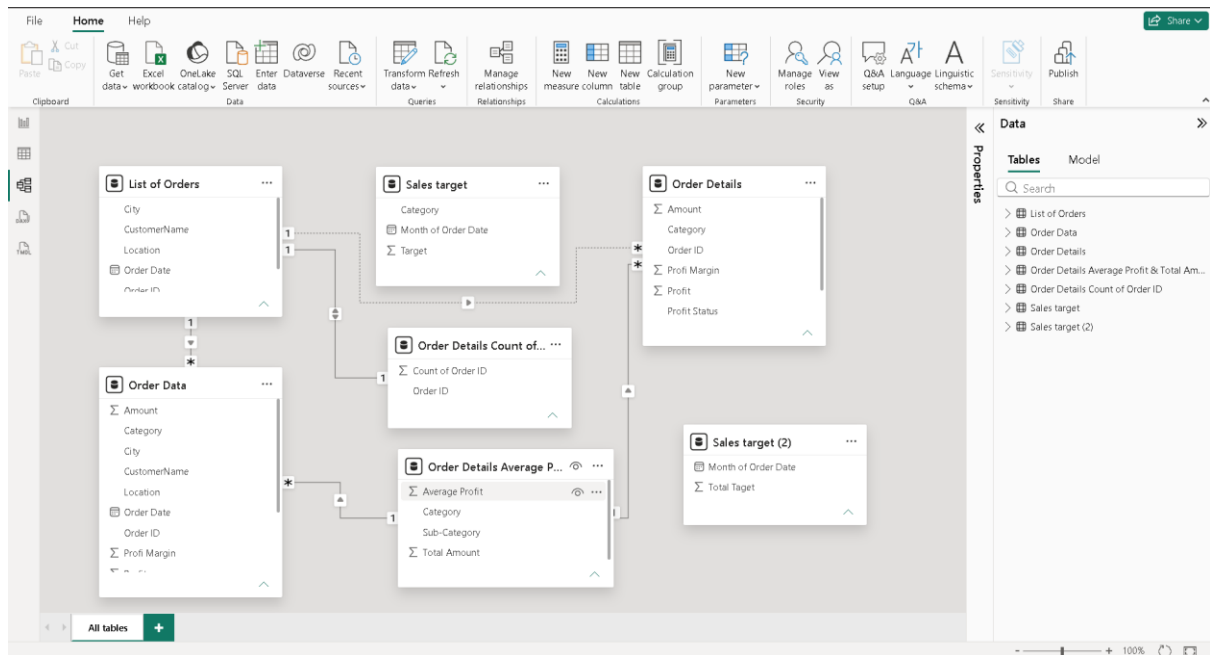
10.1 Relationships Created

- List of Orders (Order ID) → Order Details (Order ID)

o One-to-Many (1:)*

- Order Details (Category) → Sales Target (Category)

o Many-to-One (:1) *



10.2 Relationship Validation

- Verified all relationships using Manage Relationships.
- Ensured both relationships are Active and correctly defined.

Insights & Observations:

Based on grouped and aggregated data:

- Electronics shows the highest average profit.
- Furniture has moderate profitability but high order volume.
- Clothing has lower profit margins.
- Top sub-categories contribute significantly to total revenue.
- Monthly targets vary, enabling performance comparison.
- Tamil Nadu shows strong order activity in filtered analysis.

Conclusion:

This project successfully demonstrates how raw CSV files can be transformed into a structure analytical model using Power Query and Power BI. Through data cleaning, formatting, merging, aggregation, and modelling, a reliable dataset was created for reporting and visualization. The final model supports category-wise profit analysis, monthly target comparison, and state-level filtering, enabling meaningful business insights.